

Data Analytics Internship

Project 2: Exploratory Data Analysis (EDA)

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Goal

Analyze the cleaned e-commerce sales dataset to understand patterns, trends, and distributions across 1,200 order records, using descriptive statistics and outlier detection.

Key Metrics

Metric	Value	Context
Average Quantity Ordered	2.95	Units per order
Average Unit Price	\$356.41	Mean catalog price
Median Unit Price	\$364.21	50th percentile price
Total Number of Orders	1,200	Total transactions
Highest TotalPrice	\$3,456.40	Maximum order value
Lowest TotalPrice	\$11.39	Minimum order value
Top Product by Revenue	Chair	Revenue: \$195,620.11
Top Product by Order Count	Printer	Orders: 181
Most Frequent Payment Method	Online	Transactions: 258

Statistical Outlier Analysis

Statistical Metric	Value	Description
Mean TotalPrice	\$1,053.97	Average order value
Median TotalPrice	\$823.62	50th percentile order value
Standard Deviation	\$819.86	Order value dispersion
Q1 (25th Percentile)	\$410.52	25% of orders below
Q3 (75th Percentile)	\$1,578.48	75% of orders below
Interquartile Range (IQR)	\$1,167.96	Spread of middle 50%
Upper IQR Fence (Q3 + 1.5xIQR)	\$3,330.41	Tukey outlier cutoff
Lower IQR Fence (Q1 - 1.5xIQR)	-\$1,341.41	No low outliers
Identified High Outliers	8	Orders above IQR fence

All 8 high outliers involved maximum basket sizes (5 units) of premium catalog products (Tablet, Monitor, Laptop, Chair, Printer), peaking at \$3,456.40. No low-value outliers exist, since the minimum order recorded is \$11.39.

Distribution Comparison: The dataset is right-skewed. Mean TotalPrice (\$1,053.97) exceeds Median TotalPrice (\$823.62) by \$230.35, indicating that a smaller number of high-value orders are pulling the average upward, while most individual orders fall below that average.

Yearly Trend Breakdown

Year	Order Count	Total Revenue
2023	510	\$552,643.24
2024	459	\$480,235.87
2025	231	\$231,882.85

Order volume and revenue both decline year over year from 2023 to 2025. Note that 2025 data covers a partial year within the dataset's timeframe, which contributes to its lower totals.

Key Observations

1. Chair leads in revenue, Printer leads in volume. Chair generated the highest total revenue (\$195,620.11) despite Printer having the most orders (181), meaning Chairs carry a higher average order value and are more profitable per transaction.
2. Online payment dominates. Online was the most frequent payment method with 258 transactions, suggesting customers favor digital payment over cash or card options.
3. Order values are right-skewed. Mean TotalPrice (\$1,053.97) is notably higher than Median TotalPrice (\$823.62), a difference of \$230.35. This shows a smaller group of high-value orders is pulling the average up, while most orders sit below it.
4. Eight high-value outliers identified. Using the IQR method, 8 orders exceeded the statistical cutoff of \$3,330.41. All involved maximum basket sizes (5 units) of premium products, indicating genuine bulk purchases rather than data errors.
5. No low-value outliers exist. The lowest order recorded was \$11.39, which falls within the expected range, meaning there are no suspiciously low or corrupted entries in the pricing data.
6. Orders and revenue declined year over year. 2023 recorded 510 orders and \$552,643.24 in revenue, dropping to 459 orders and \$480,235.87 in 2024, and 231 orders and \$231,882.85 in 2025 (partial year of data).

Conclusion

This analysis reveals that while Printers drive order volume, Chairs are the true revenue leader, showing that product popularity and profitability don't always align. The right-skewed distribution of order values, with the mean sitting above the median, confirms that a small group of high-value orders shapes the average more than typical customer behavior does. Customer behavior leans heavily toward online payments, and the 8 identified outliers are not data errors but genuine high-value purchases, most likely bulk buyers who should be considered for loyalty or premium customer programs. The year-over-year decline in both order volume and revenue from 2023 to 2025 is worth flagging for further investigation, though the 2025 figures reflect only a partial year within the dataset. Overall, the dataset shows healthy variation in order values with no signs of corrupted or unusually low-value transactions, confirming the dataset's reliability going into deeper analysis.

Tools Used

Excel, Gemini AI (within Google Sheets for statistical calculations), descriptive statistics, IQR-based outlier detection.